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1 C:\Users\sumer\Desktop\Projects\Python\test\venv\Scripts\python
  .exe C:\Users\sumer\Desktop\Projects\Python\test\DecisionTree.
  py
2 Classification report for all hyperparameter combinations:
3 Hyperparameters: {'max_depth': 5, 'min_samples_split': 2}
4 Mean test score: 0.819 (std: 0.003)
5 Classification report:
6           precision    recall  f1-score   support
7
8      0           0.84      0.94      0.89       7040
9      1           0.64      0.37      0.47       1960
10
11     accuracy                0.82       9000
12    macro avg           0.74      0.66      0.68       9000
13 weighted avg           0.80      0.82      0.80       9000
14
15 =====
16 =====
17 Hyperparameters: {'max_depth': 5, 'min_samples_split': 5}
18 Mean test score: 0.819 (std: 0.003)
19 Classification report:
20           precision    recall  f1-score   support
21
22      0           0.84      0.94      0.89       7040
23      1           0.64      0.37      0.47       1960
24
25     accuracy                0.82       9000
26    macro avg           0.74      0.66      0.68       9000
27 weighted avg           0.80      0.82      0.80       9000
28
29 =====
30 =====
31 Hyperparameters: {'max_depth': 5, 'min_samples_split': 10}
32 Mean test score: 0.819 (std: 0.003)
33 Classification report:
34           precision    recall  f1-score   support
35
36      0           0.84      0.94      0.89       7040
37      1           0.64      0.37      0.47       1960
38
39     accuracy                0.82       9000
40    macro avg           0.74      0.66      0.68       9000
41 weighted avg           0.80      0.82      0.80       9000
42
43 =====
44 =====
45 Hyperparameters: {'max_depth': 10, 'min_samples_split': 2}
46 Mean test score: 0.805 (std: 0.005)

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44 Classification report:

		precision	recall	f1-score	support
47	0	0.84	0.94	0.89	7040
48	1	0.64	0.37	0.47	1960
50	accuracy			0.82	9000
51	macro avg	0.74	0.66	0.68	9000
52	weighted avg	0.80	0.82	0.80	9000

53

54 =====
 =====

55 Hyperparameters: {'max_depth': 10, 'min_samples_split': 5}

56 Mean test score: 0.805 (std: 0.006)

57 Classification report:

		precision	recall	f1-score	support
60	0	0.84	0.94	0.89	7040
61	1	0.64	0.37	0.47	1960
63	accuracy			0.82	9000
64	macro avg	0.74	0.66	0.68	9000
65	weighted avg	0.80	0.82	0.80	9000

66

67 =====
 =====

68 Hyperparameters: {'max_depth': 10, 'min_samples_split': 10}

69 Mean test score: 0.805 (std: 0.004)

70 Classification report:

		precision	recall	f1-score	support
73	0	0.84	0.94	0.89	7040
74	1	0.64	0.37	0.47	1960
76	accuracy			0.82	9000
77	macro avg	0.74	0.66	0.68	9000
78	weighted avg	0.80	0.82	0.80	9000

79

80 =====
 =====

81 Hyperparameters: {'max_depth': 15, 'min_samples_split': 2}

82 Mean test score: 0.783 (std: 0.006)

83 Classification report:

		precision	recall	f1-score	support
86	0	0.84	0.94	0.89	7040
87	1	0.64	0.37	0.47	1960

88

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89      accuracy                0.82      9000
90      macro avg              0.74      0.66      0.68      9000
91      weighted avg           0.80      0.82      0.80      9000
92
93 =====
94      Hyperparameters: {'max_depth': 15, 'min_samples_split': 5}
95      Mean test score: 0.782 (std: 0.008)
96      Classification report:
97              precision      recall  f1-score      support
98
99              0              0.84      0.94      0.89      7040
100             1              0.64      0.37      0.47      1960
101
102      accuracy                0.82      9000
103      macro avg              0.74      0.66      0.68      9000
104      weighted avg           0.80      0.82      0.80      9000
105
106 =====
107      Hyperparameters: {'max_depth': 15, 'min_samples_split': 10}
108      Mean test score: 0.783 (std: 0.008)
109      Classification report:
110              precision      recall  f1-score      support
111
112              0              0.84      0.94      0.89      7040
113             1              0.64      0.37      0.47      1960
114
115      accuracy                0.82      9000
116      macro avg              0.74      0.66      0.68      9000
117      weighted avg           0.80      0.82      0.80      9000
118
119 =====
120      Best hyperparameters found: {'max_depth': 5, '
121      min_samples_split': 10}
122      Decision Tree Classification Best Accuracy: 0.8173333333333334
123
124      Process finished with exit code 0
125
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